CORE JAVA

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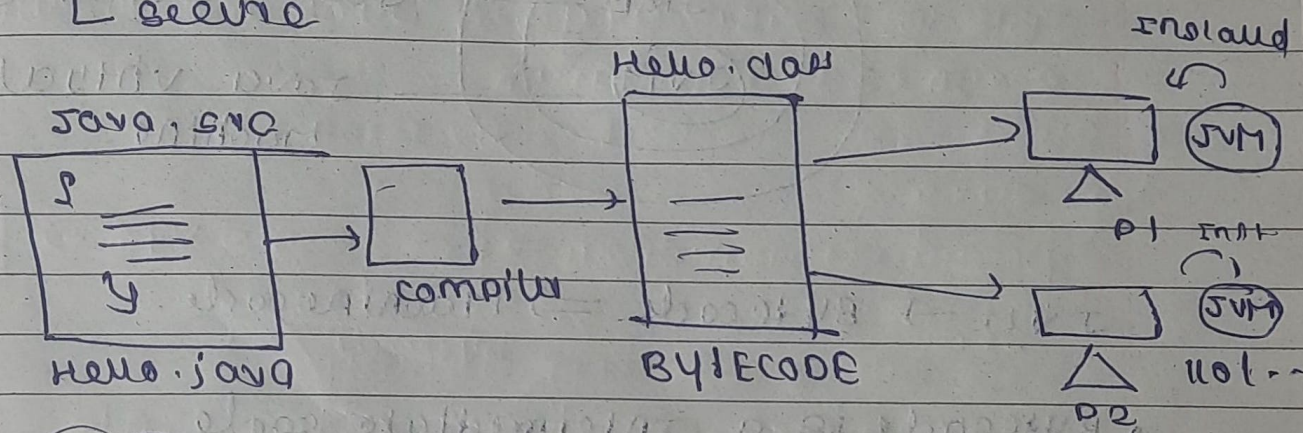
Why Java ?

Java solved platform dependent problem of C & C++

C / C++
↳ platform dependent
↳ simplicity
↳ security prob
⇓
JAVA

Reason

JAVA → WORA (write once Run Anywhere)
↳ Portable
↳ simple
↳ secure

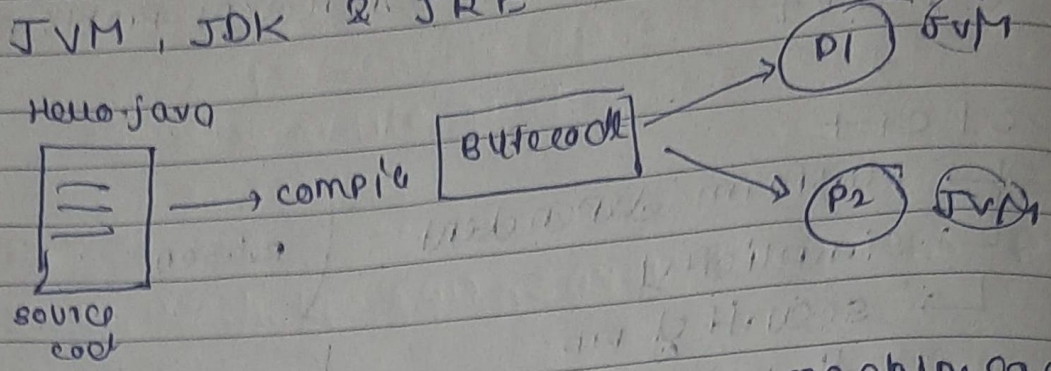


convert bytecode into machine — JVM work is to take source code & convert it into specific platform instruction
JVM platform dependent for platform.

security (data crash)
Backend → server (Programs APIs)
Frontend → Applications (run on user browser) (initial frontend) — usecase

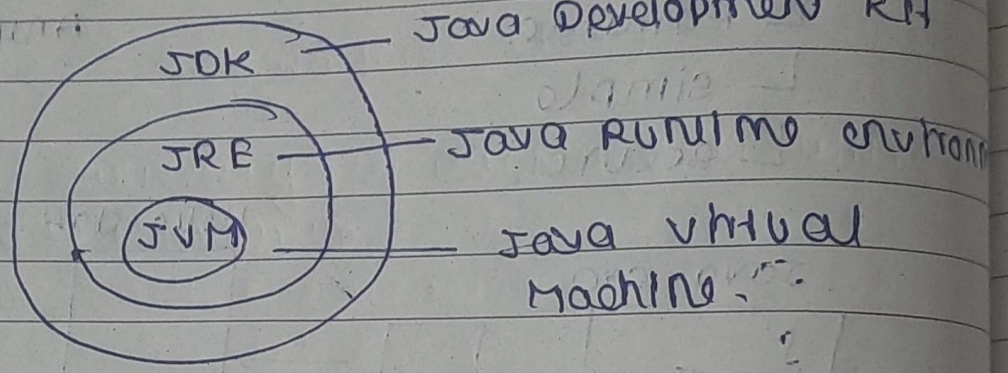
JVM provide secured environment → sandbox Model

* JVM, JDK & JRE



JVM converts the byte code into machine code.

Hierarchy



- JVM → Bytecode ⇒ Machine code.

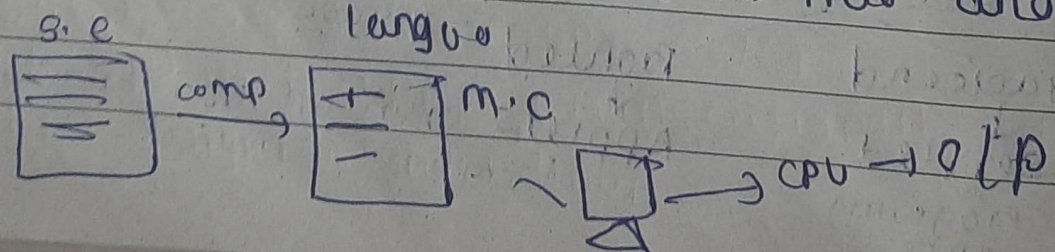
- Bytecode is a Intermediate code

software needed to convert Byte code into machine code.

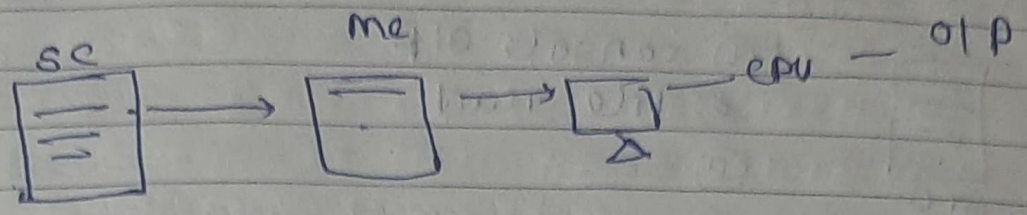
(1) compiler
(2) Interpreter
→ s.c → machine code

read at home

compiler → convert one program into another language



Interpreter
convert source code into machine code line by line.



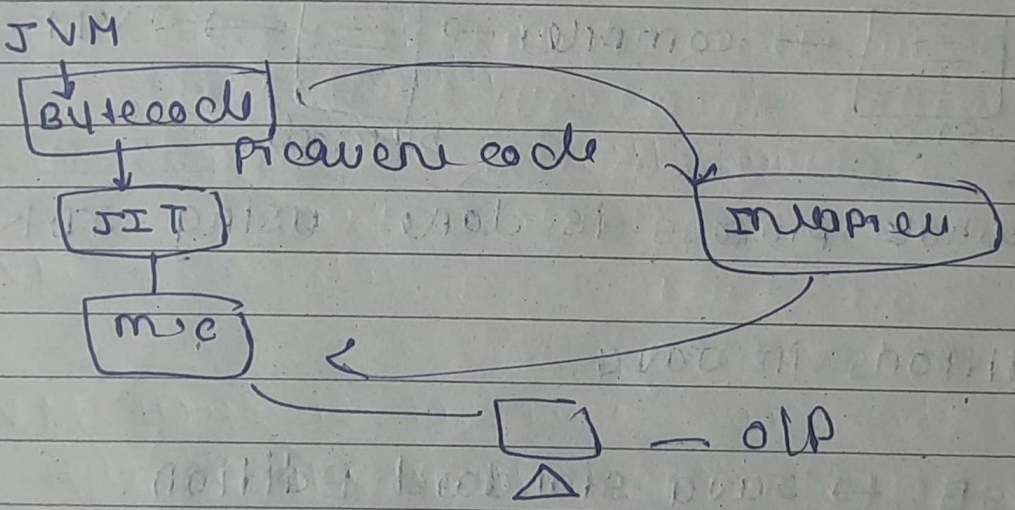
Generate line by line

Interpreter
↳ convert code line by line

compiler
↳ convert code all at once.

* Java is both compiled & Interpreted

Interpreter + JIT compiler
(Just-in-time)



JVM
↳ ① convert Bytecode to m.c (Interpreter + JIT compiler)
② security (sand-box model)
③ Garbage collection

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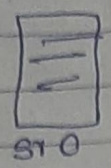
2) JRE — inbuilt run

↳ JVM + [class libraries] — all operation done internally



→ ① console o/p
② file read
③ ≡

JRE used to run java program properly

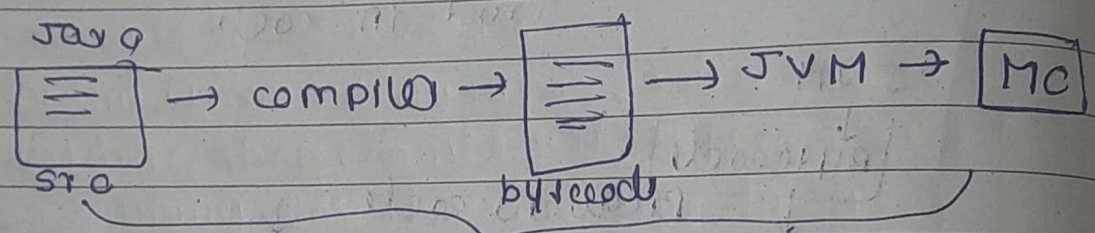
Then we come to JDK

3)* JDK (Java Development Kit)

↳ [JRE + compiler
debugger
javadocs etc]

JDK is basically a package to write & run java program.

↳



whole process is done using JDK

4 Editions in Java

JSE → Java Standard Edition
JEE → Java Enterprise Edition
JME → Java Micro Edition

→ JSE = core Java, OOPS ... etc
→ JEE = → web apps (transcational) = JakartaEE
→ JME = light weight Edition